



System Design Report

for

Virtual Fitness Trainer

Version 1.0

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0.1 Introduction

This document presents the comprehensive system design for the **Virtual Fitness Trainer** project, as per the requirements specified in the approved Software Requirements Specification (SRS) Version 2.0 dated December 28, 2025. The purpose of this milestone is to translate documented functional and non-functional requirements into a concrete system design using standard software modeling techniques.

The **Virtual Fitness Trainer** is a web-based platform designed to provide personalized fitness and diet guidance through AI-powered plan generation, progress tracking, and instructional video demonstrations. This design report aligns with the objectives of Project Milestone 3, ensuring traceability between requirements and design artifacts, validating system behavior and structure, and facilitating stakeholder feedback through prototyping.

0.2 Design Assumptions and Constraints

0.2.1 Assumptions

The following assumptions were made during the system design, based on the SRS:

1. Users have access to a modern web browser and a stable internet connection.
2. AI/ML models for workout and diet plan generation are pre-trained and available via a service endpoint.
3. Users will provide accurate health and fitness information for effective personalization.
4. Workout demonstration videos are pre-recorded and stored in a media repository.
5. The system will be deployed on a cloud platform (Vercel) with MongoDB Atlas as the database service.
6. Users with serious medical conditions will consult healthcare professionals before following generated plans.

0.2.2 Constraints

Design constraints derived from the SRS include:

1. Technology Stack:

- Web interface: Python-based framework (e.g., Flask/Django)
- Backend: Node.js with Express.js
- Database: MongoDB 5.0+
- AI Framework: TensorFlow/PyTorch

2. Performance:

- Login response time: 2 seconds

- Workout plan generation 5 seconds
- Support for 1000 concurrent users

3. Security:

- Passwords hashed with bcrypt (min. 10 rounds)
- HTTPS/TLS 1.2+ for all communications
- JWT-based authentication with token expiration
- Input sanitization against SQL/NoSQL and XSS attacks

4. Compliance:

- Adherence to health data privacy regulations
- OWASP Top 10 security standards
- Accessibility standards (WCAG 2.1 Level AA)

5. Availability: 99.5% uptime during operational hours, excluding scheduled maintenance.

0.3 Key Design Decisions

0.3.1 Decomposition in Data Flow Diagrams

- The AI-based plan generation (FR-3, FR-4) is decomposed into sub-processes: *User Data Retrieval*, *Model Input Preparation*, *AI Model Execution*, *Result Filtering and Scheduling*, and *Plan Storage* (see Level 2 DFD in Figure 3.5). This ensures modularity and aligns with the SRS processing steps.
- User registration (FR-1) is broken down into *Input Validation*, *Email Uniqueness Check*, *Password Hashing*, *Age/BMI Calculation*, and *Database Storage* (shown in DFD Level 1) to isolate concerns and improve maintainability.

0.3.2 Class Relationships in Class Diagram

- **Association:** User *has a* Profile, User *generates* WorkoutPlan and DietPlan.
- **Aggregation:** ProgressTracking *contains* multiple ProgressEntry objects.
- **Composition:** WorkoutPlan *is composed of* Exercise objects; DietPlan *is composed of* Meal objects.
- **Inheritance:** User class extended to *RegisteredUser* and *Trainer* (for future role expansion).

0.3.3 Distribution of Functionality Across Diagrams

- Separate sequence diagrams are created for each functional requirement (FR-1 to FR-9) to avoid overcrowding and improve clarity (Figures 3.6 to 3.10).
- Activity diagrams are created for key workflows: *User Login and Registration* (Figure 3.11) and *Core System Activities* including Workout Plan Generation (Figure 3.12).
- Two component diagrams illustrate the architecture: high-level system components (Figure 3.14) and detailed AI/Recommendation subsystem (Figure 3.15), reflecting a microservices-oriented approach.

0.4 System Design Diagrams

0.4.1 Use Case Diagram

The use case diagram identifies primary actors (User, Fitness Trainer, AI System) and major use cases derived from SRS functional requirements.

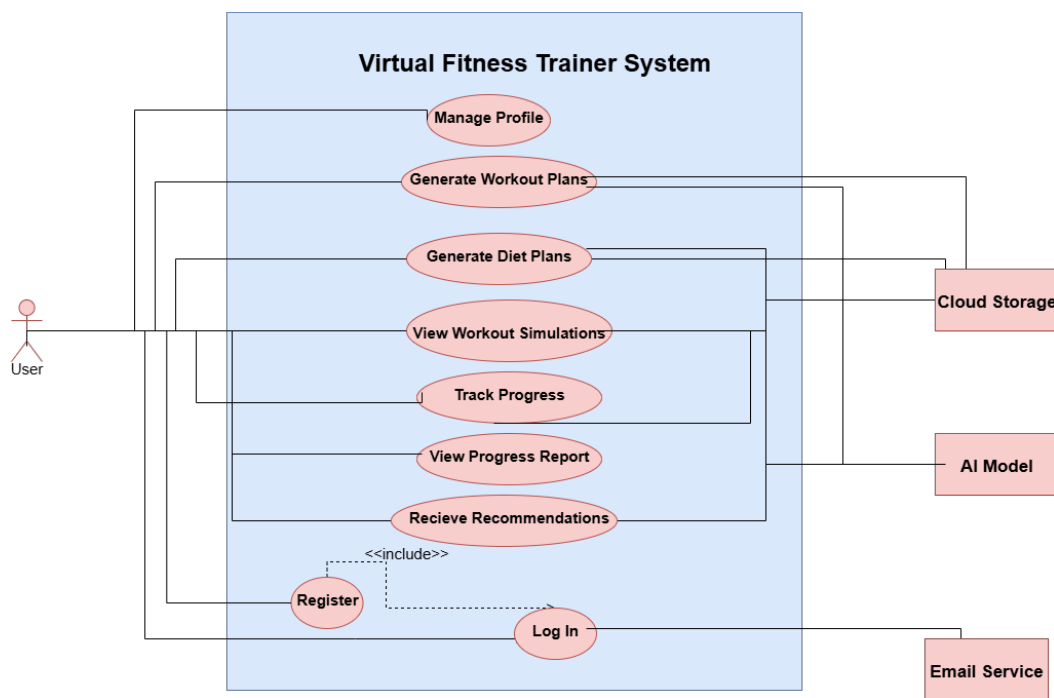


Figure 1: Use Case Diagram for Virtual Fitness Trainer

0.4.2 Data Flow Diagrams

Level 0 (Context Diagram)

Shows the system as a single process interacting with external entities: User, AI Model Service, Email Service, and Database.

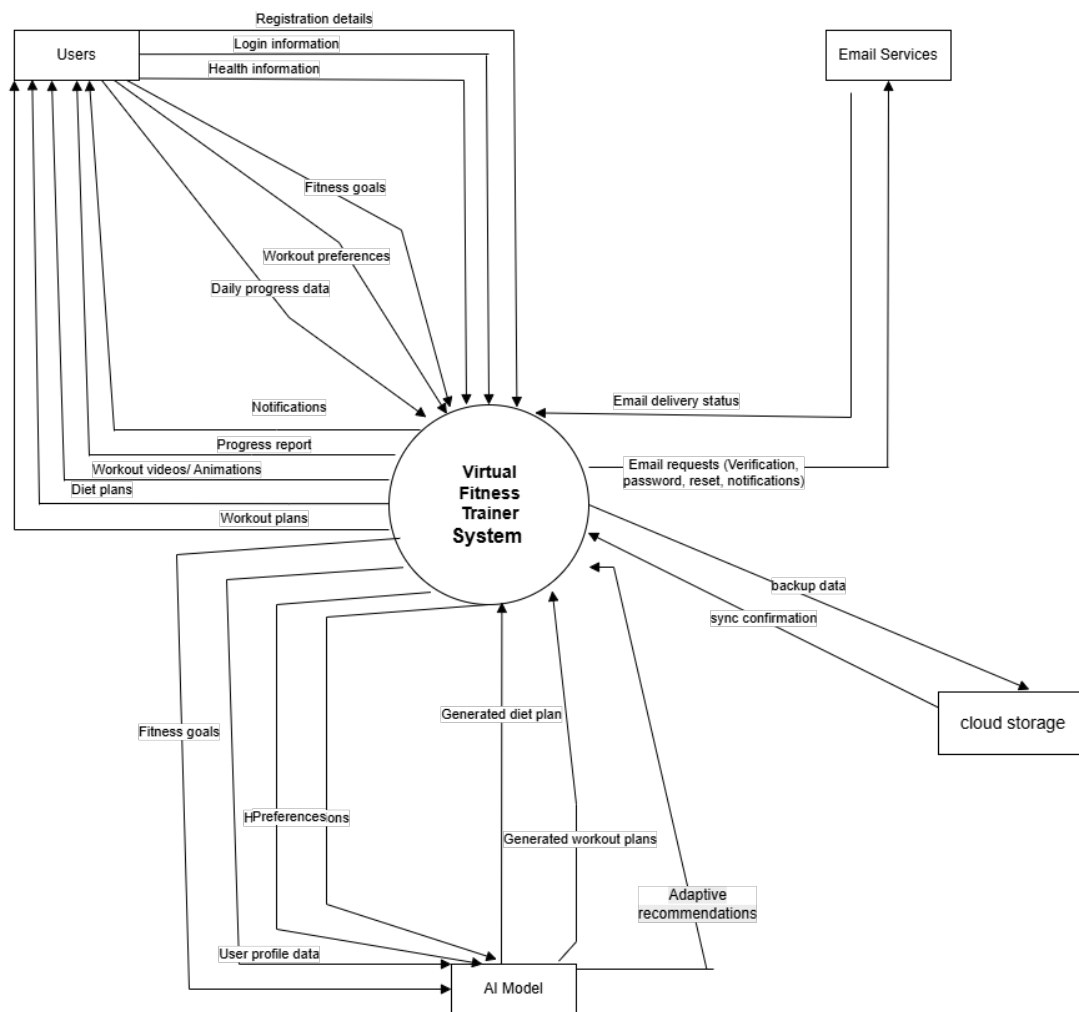


Figure 2: DFD Level 0 - Context Diagram

Level 1 DFD

Decomposes the system into major processes: User Management (Process 1.x), Plan Generation (Process 2.x), Progress Tracking (Process 4.x), and Reporting.

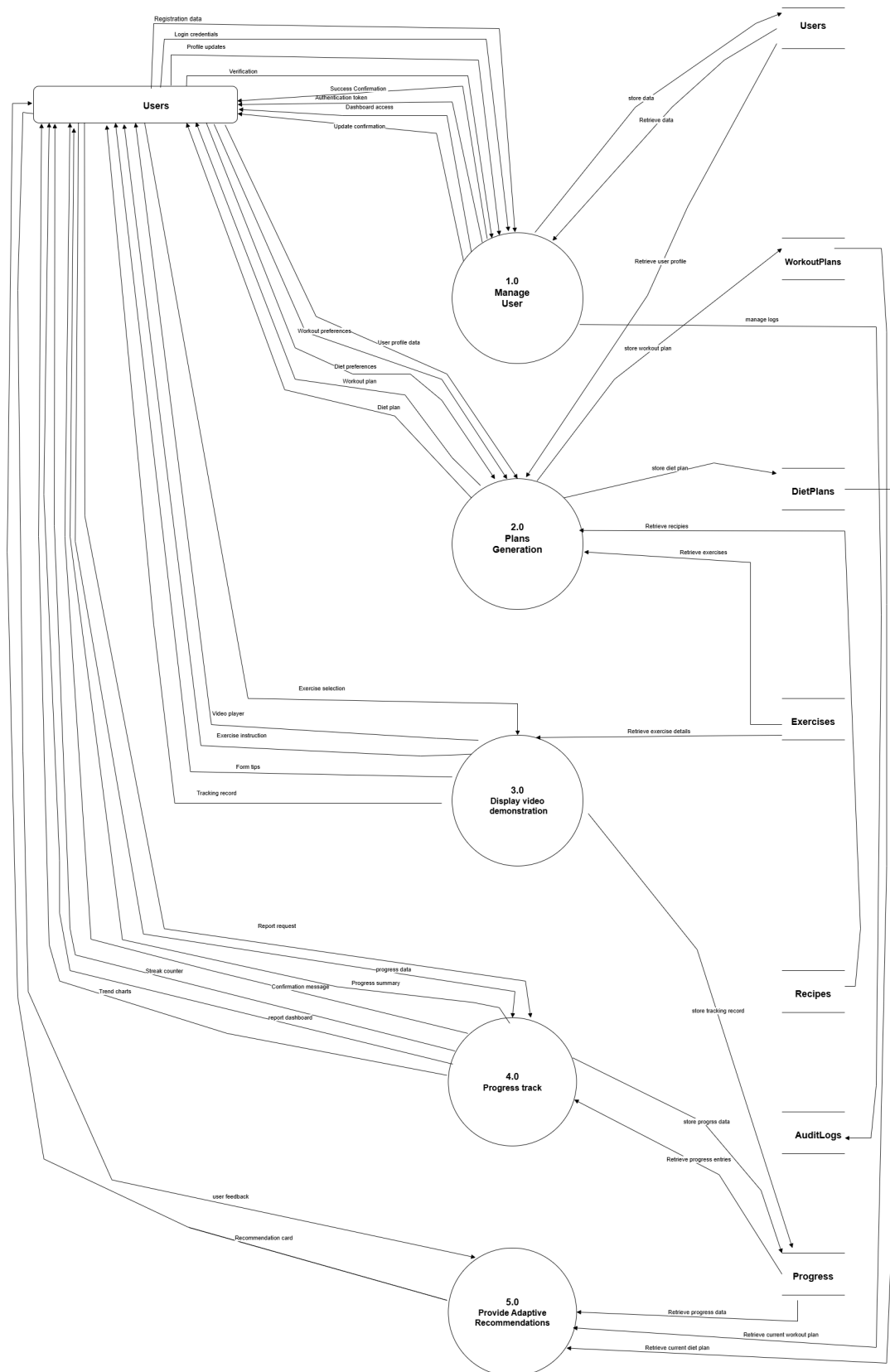


Figure 3: DFD Level 1 - Major Processes

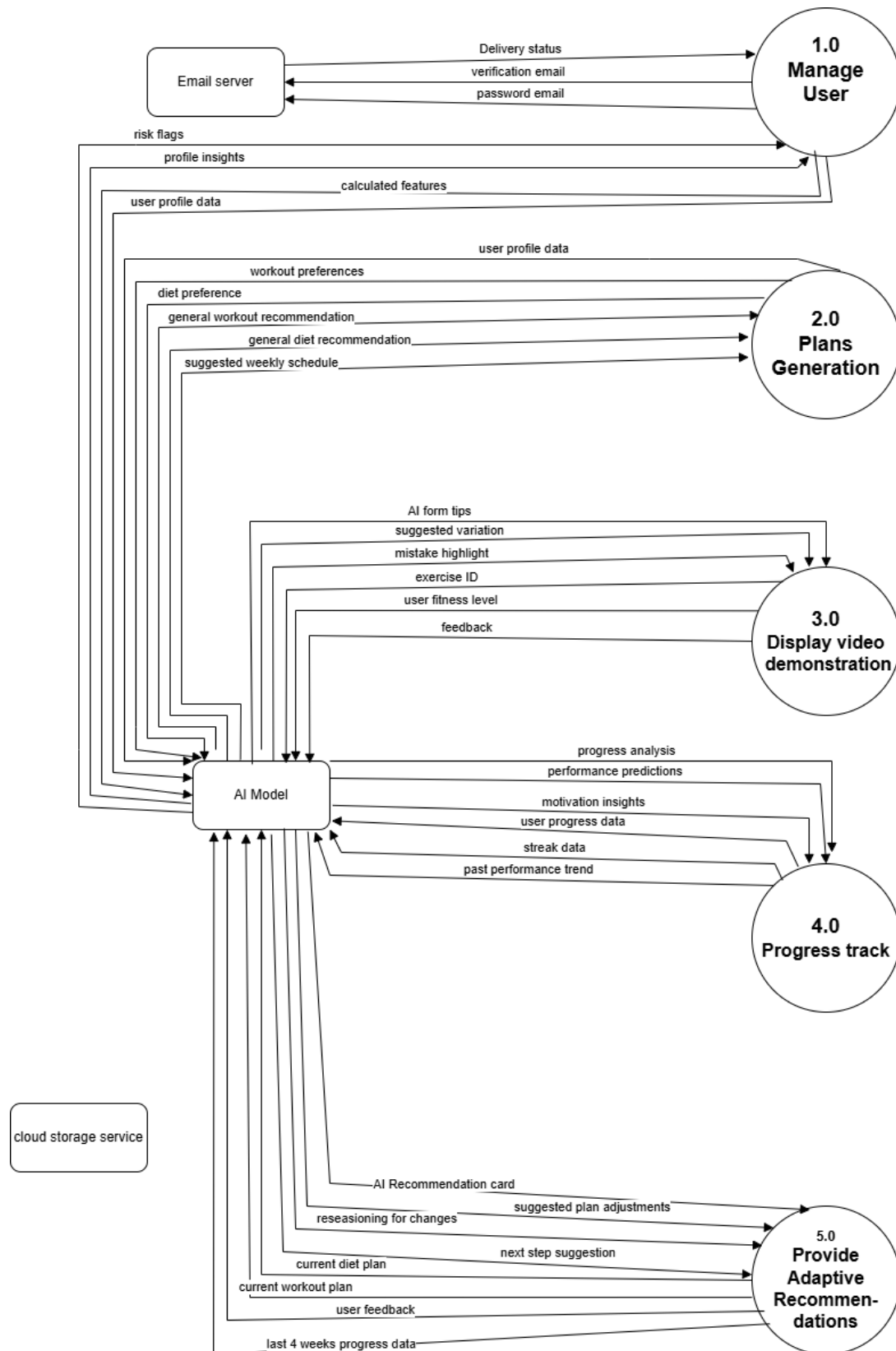


Figure 4: DFD Level 1 - Major Processes (Cont.)

Level 2 DFD (for Plan Generation)

Further decomposes the AI-based plan generation process (Process 2.1 and 2.2) into sub-processes: User Data Retrieval, Model Input Preparation, AI Model Execution, Result Filtering and Scheduling, and Plan Storage.

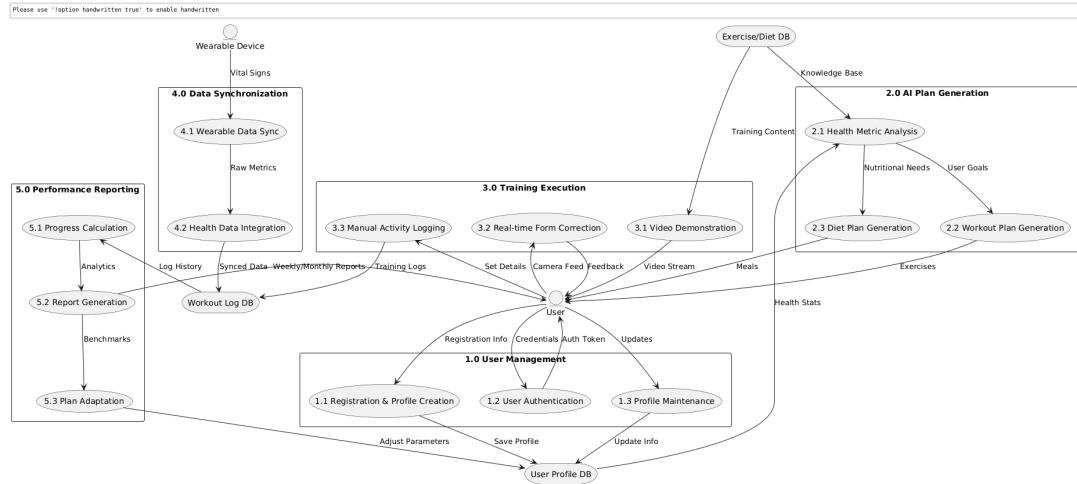


Figure 5: DFD Level 2 - Plan Generation Process

0.4.3 Sequence Diagrams

Sequence diagrams model interactions for key functionalities.

User Registration (FR-1)

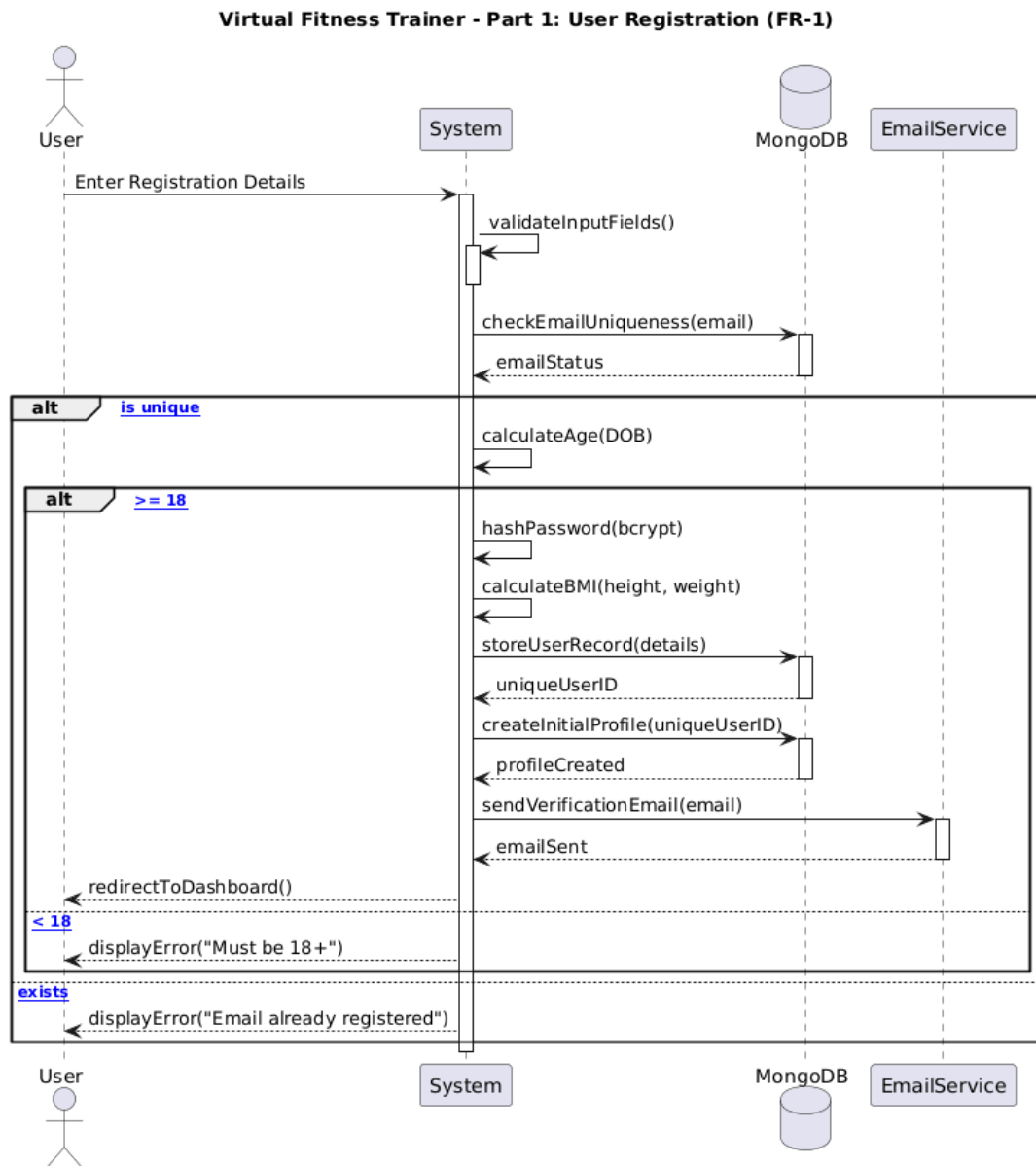


Figure 6: Sequence Diagram - User Registration

User Login (FR-2)

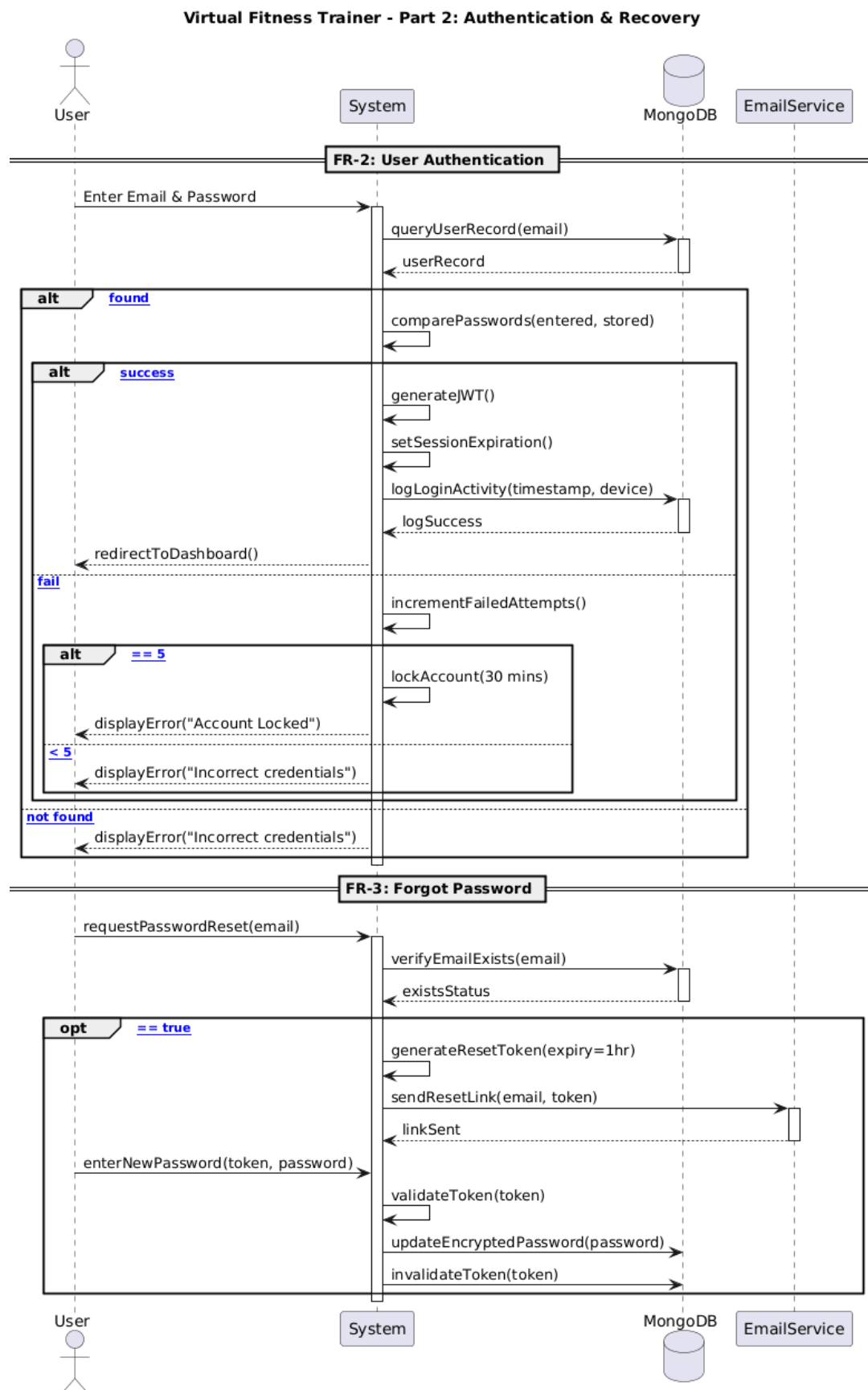
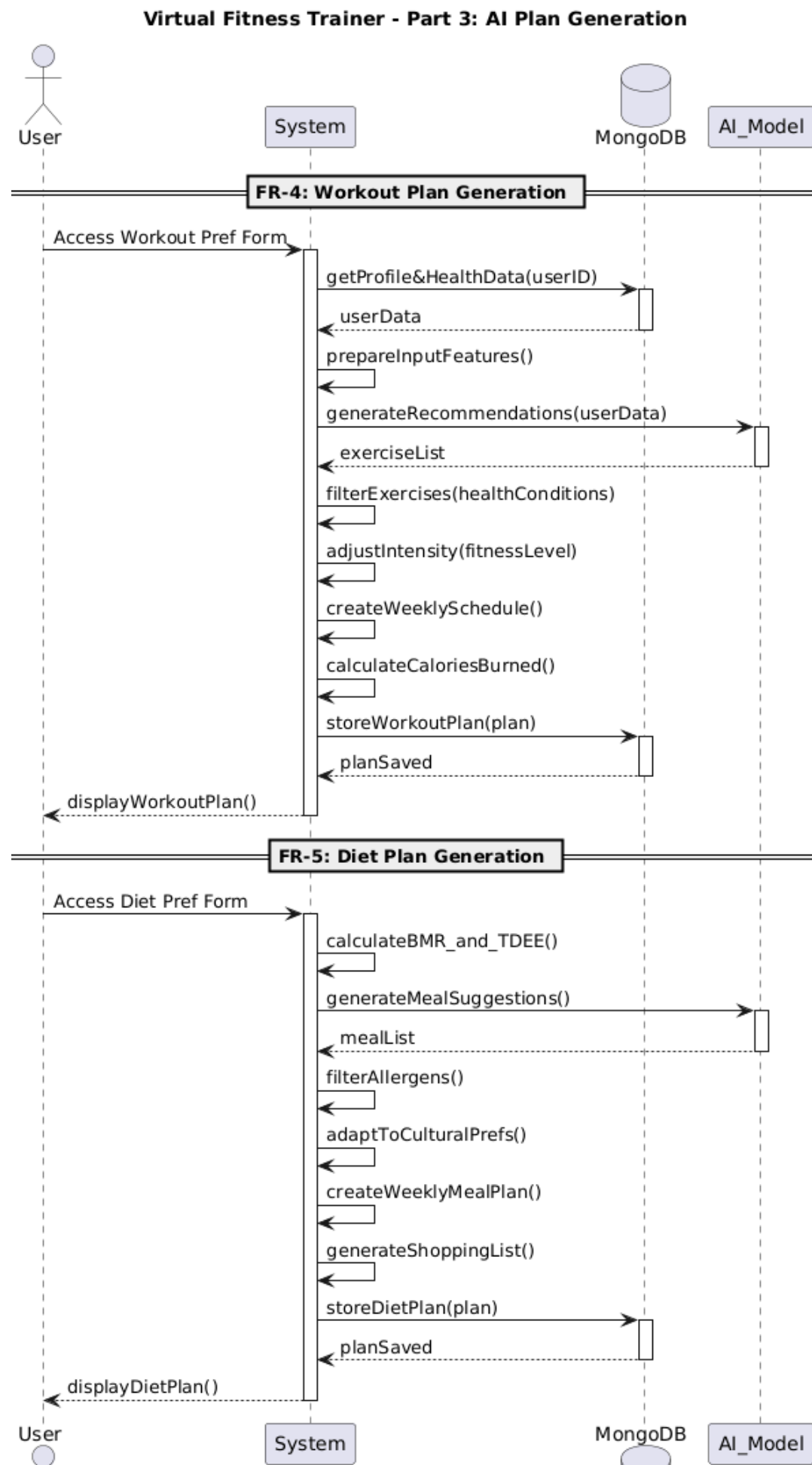


Figure 7: Sequence Diagram - User Login

Workout Plan Generation (FR-3) and Diet Plan Generation (FR-4)



Daily Progress Tracking (FR-6) and Video Demonstration (FR-5)

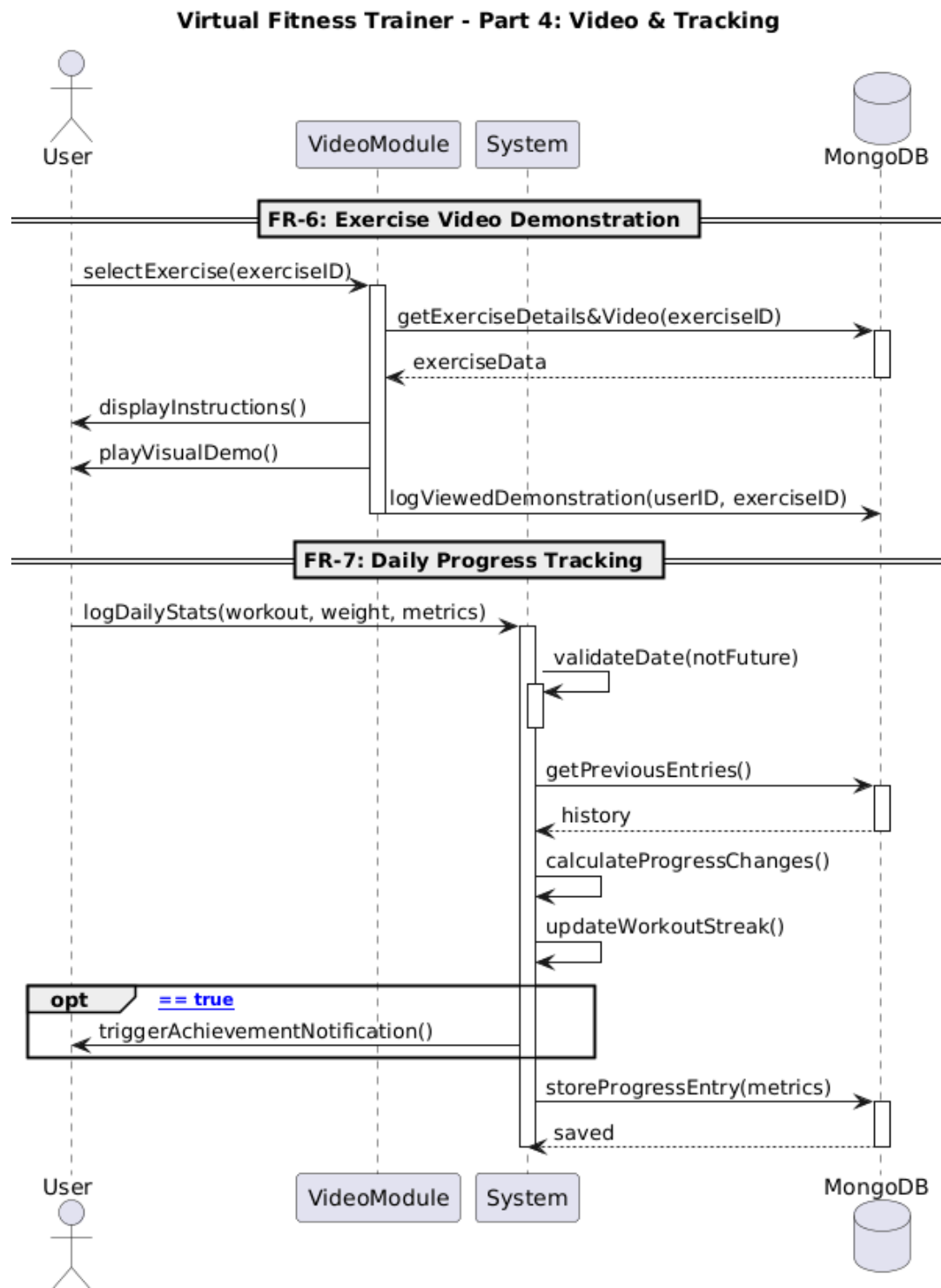


Figure 9: Sequence Diagram - Daily Progress Tracking and Video Demonstration

Progress Reports, Plan Recommendations and User Profile Management

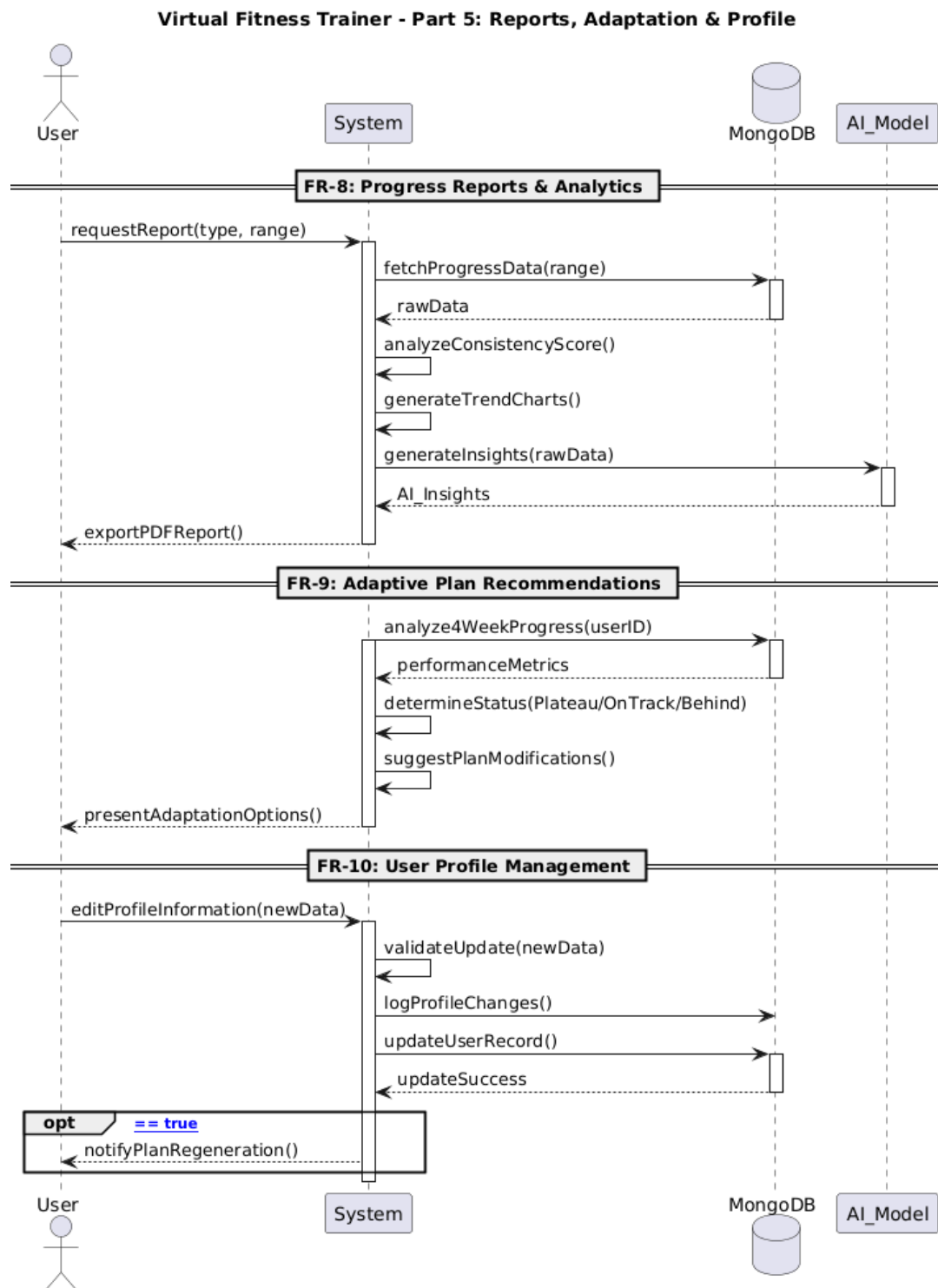


Figure 10: Sequence Diagram - Reports, Recommendations, and Profile Management

0.4.4 Activity Diagrams

Activity diagrams represent important system workflows.

User Login and Registration

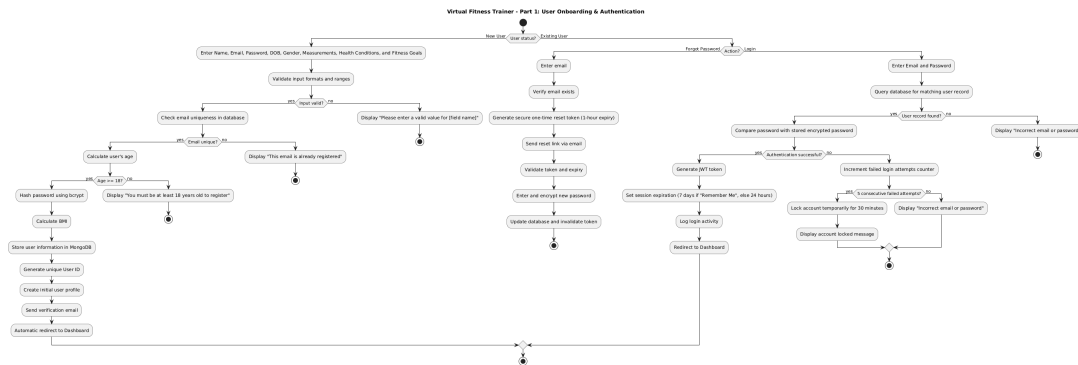


Figure 11: Activity Diagram - User Login and Registration

Core System Activities

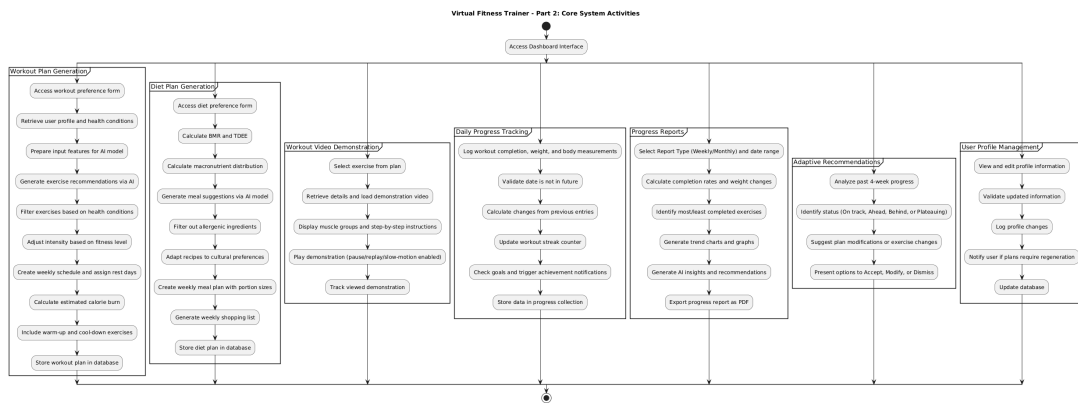


Figure 12: Activity Diagram - Workout Plan Generation

0.4.5 Class Diagram

The class diagram reflects the system structure with classes, attributes, methods, and relationships.

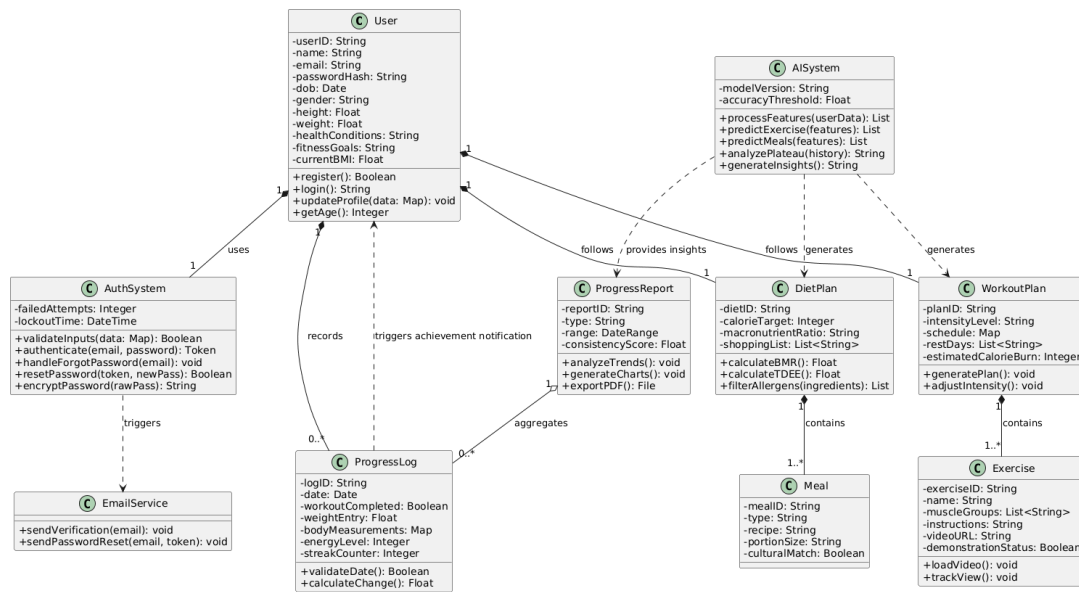


Figure 13: Class Diagram for Virtual Fitness Trainer

0.4.6 Component Diagram

Illustrates high-level system components and their dependencies.

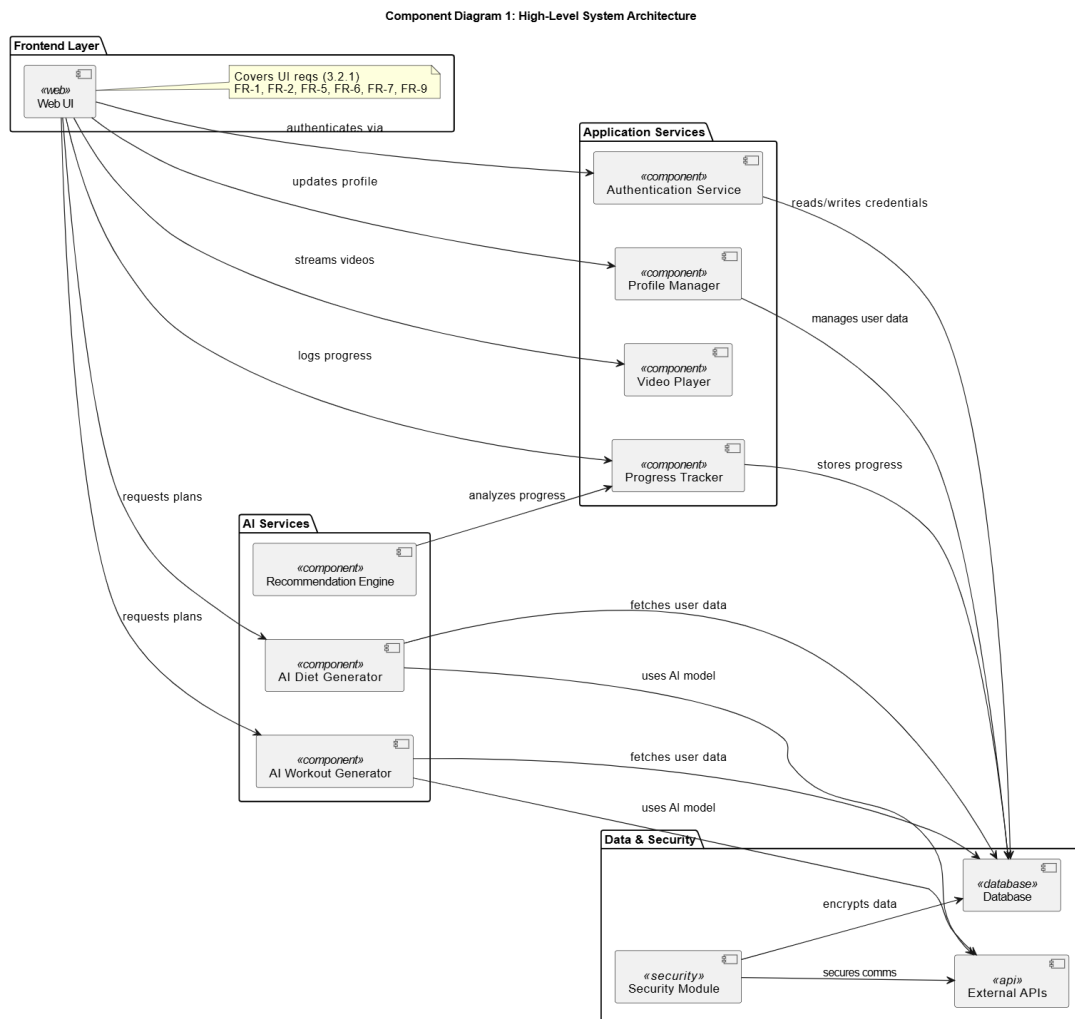


Figure 14: High Level System Architecture

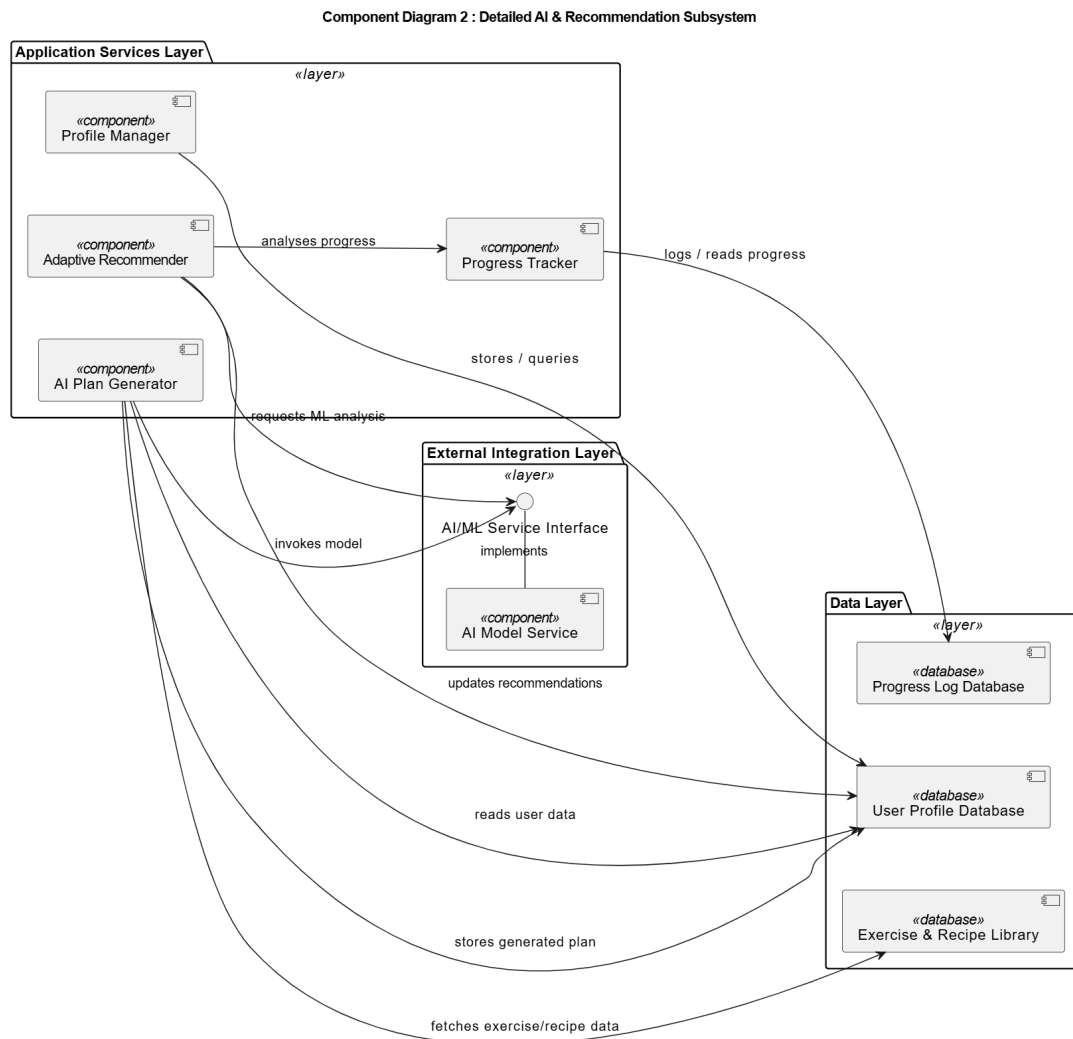


Figure 15: Detailed AI and Recommendation System

0.5 Requirements–Design Traceability Table

Req. ID	Description	Mapped Design Elements
FR-1	User Registration and Profile Creation	•Use Case: Register User •DFD Process: 1.1 User Registration •Sequence Diagram: seq 1.png •Class: User, UserProfile •Prototype: Registration Page
FR-2	User Authentication and Login	•Use Case: Login, Forgot Password •DFD Process: 1.2 Authenticate User •Sequence Diagram: seq 2.png •Class: AuthService, User •Prototype: Login Page, Password Reset
FR-3	AI-Based Workout Plan Generation	•Use Case: Generate Workout Plan •DFD Process: 2.1 Generate Workout Plan •Sequence Diagram: seq 3.png •Activity Diagram: activity part2.png •Class: WorkoutPlan, Exercise, AIService •Prototype: Workout Plan Form, Results
FR-4	AI-Based Diet Plan Generation	•Use Case: Generate Diet Plan •DFD Process: 2.2 Generate Diet Plan •Sequence Diagram: seq 3.png •Class: DietPlan, Meal, AIService •Prototype: Diet Preferences Form, Meal Plan
FR-5	Workout Video Demonstration	•Use Case: View Exercise Demo •DFD Process: 3.1 Show Video Demonstration •Sequence Diagram: seq 4.png •Class: Exercise, VideoService •Prototype: Video Player Page
FR-6	Daily Progress Tracking	•Use Case: Log Daily Progress •DFD Process: 4.1 Track Progress •Sequence Diagram: seq 4.png •Class: ProgressTracking, ProgressEntry •Prototype: Daily Log Page
FR-7	Weekly/Monthly Progress Reports	•Use Case: Generate Report •DFD Process: 4.2 Generate Reports •Sequence Diagram: seq 5.png •Class: ReportGenerator, ProgressTracking •Prototype: Reports Dashboard
FR-8	Adaptive Plan Recommendations	•Use Case: Receive Recommendations •DFD Process: 2.3 Adapt Plan •Sequence Diagram: seq 5.png •Class: RecommendationEngine, WorkoutPlan, DietPlan •Prototype: Recommendations Panel
FR-9	User Profile Management	•Use Case: Update Profile •DFD Process: 1.3 Manage Profile •Sequence Diagram: seq 5.png •Class: UserProfile •Prototype: Profile Settings Page

Table 1: Traceability Matrix: Functional Requirements to Design Artifacts

Traceability Matrix: Functional Requirements to Design Artifacts

0.5.1 Repository and Prototype Links

- **GitHub Repository:** <https://github.com/wasif-cmd/Virtual-Fitness-Trainer>
- **Figma Prototype:** <https://www.figma.com/design/fBCSFfLgw5XEsQDaN90pt1/Virtual-Fitness-Trainer?node-id=0-1&t=rS5Bl7JZMQHveF1R-1>